

Questions

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Polynomials

1. For a real constant k , consider the equation

$$x^4 - 4x^2 + k = 0$$

Which of the following conditions on k is **necessary and sufficient** for the equation to have exactly two distinct real solutions?

- A $k > 4$
- B $k = 4$
- C $k \leq 0$
- D $0 < k < 4$
- E $k = 0$
- F $k < 0$ or $k = 4$
- G $k < 0$

Number

2. Let n be an integer. Consider the following conditions:

- I:** n is odd
- II:** n is a multiple of 4
- III:** n is a prime greater than 3

Which of these conditions is/are **sufficient** to guarantee that $n^3 - n$ is divisible by 12?

- A** None of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II, and III

Polynomials

- 3.** The curve C has equation $y = x^3 - 3x^2$

A tangent to C passes through the point $P(0, 1)$. The point at which the tangent touches C has x -coordinate t

Find the complete set of possible values of t

A $t = -\frac{1}{2}$ only

B $t = 1$ only

C $t = -\frac{1}{2}$ or $t = 1$

D $t = -1$ or $t = 1$

E $t = \frac{1}{2}$ or $t = 1$

F $t = -\frac{1}{2}$ or $t = \frac{3}{2}$

4. A student is required to solve the equation

$$3\sqrt{x+1} = x - 1$$

Their attempt is as follows:

$$3\sqrt{x+1} = x - 1 \xrightarrow{\text{(I)}} 9(x+1) = (x-1)^2$$

$$9(x+1) = (x-1)^2 \xrightarrow{\text{(II)}} 9x+9 = x^2 - 2x + 1$$

$$9x+9 = x^2 - 2x + 1 \xrightarrow{\text{(III)}} x^2 - 11x - 8 = 0$$

$$x^2 - 11x - 8 = 0 \xrightarrow{\text{(IV)}} x = \frac{11 \pm 3\sqrt{17}}{2}$$

They then conclude that the solutions of the original equation are $x = \frac{11 + 3\sqrt{17}}{2}$ and $x = \frac{11 - 3\sqrt{17}}{2}$. Which one of the following is true?

- A** The solution is correct.
- B** Only one of $x = \frac{11 + 3\sqrt{17}}{2}$ and $x = \frac{11 - 3\sqrt{17}}{2}$ is correct and the error arises as a result of step (I).
- C** Only one of $x = \frac{11 + 3\sqrt{17}}{2}$ and $x = \frac{11 - 3\sqrt{17}}{2}$ is correct and the error arises as a result of step (II).
- D** Only one of $x = \frac{11 + 3\sqrt{17}}{2}$ and $x = \frac{11 - 3\sqrt{17}}{2}$ is correct and the error arises as a result of step (III).
- E** There is another value of x that satisfies the original equation and the error arises as a result of step (I).
- F** There is another value of x that satisfies the original equation and the error arises as a result of step (II).
- G** There is another value of x that satisfies the original equation and the error arises as a result of step (III).

Trigonometry

5. Let p be a real constant. Consider the equation

$$\sin x \cos x = p$$

on $0 \leq x < 2\pi$

Which of the following statements is/are true?

- I:** If $|p| < \frac{1}{2}$ then the equation has exactly 4 distinct solutions
- II:** If the equation has exactly 2 distinct solutions, then $|p| = \frac{1}{2}$

- A** None of them
- B** I only
- C** II only
- D** I and II

Logic

6. We say a sequence of real numbers x_n is a *Cauchy sequence* if

for all $\varepsilon > 0$ there exists a positive integer $N \in \mathbb{N}$ such that
for all $n, m \geq N$ we have $|x_n - x_m| < \varepsilon$

Which one of the following statements is true if and only if x_n is **not** a Cauchy sequence?

- A For every $\varepsilon > 0$ and every $N \in \mathbb{N}$ there exists $m, n \geq N$ with $|x_m - x_n| < \varepsilon$
- B There exists $\varepsilon > 0$ such that for every $N \in \mathbb{N}$ and for all $m, n \leq N$ we have $|x_m - x_n| \geq \varepsilon$
- C For every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ and there exists $m, n \geq N$ with $|x_m - x_n| \geq \varepsilon$
- D For every $N \in \mathbb{N}$ there exists $\varepsilon > 0$ such that for all $m, n \geq N$ we have $|x_m - x_n| \geq \varepsilon$
- E There exists $\varepsilon > 0$ such that for every $N \in \mathbb{N}$ there exist $m, n \geq N$ with $|x_m - x_n| \geq \varepsilon$
- F There exists $\varepsilon > 0$ such that for every $N \in \mathbb{N}$ there exist $m, n \geq N$ with $|x_m - x_n| < \varepsilon$
- G There exists $N \in \mathbb{N}$ such that for all $\varepsilon > 0$ and all $m, n \geq N$ we have $|x_m - x_n| \geq \varepsilon$
- H There exists $\varepsilon > 0$ and there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$ we have $|x_m - x_n| \geq \varepsilon$

Integration

7. For a real constant k , define

$$A(k) = \int_{-1}^2 |x - k| \, dx$$

For which value of k is $A(k)$ minimised?

A -1

B $\frac{1}{2}$

C 1

D $\frac{3}{2}$

E 2

F 0

8. Consider the following problem for an integer n :

$$\left(\frac{1}{3}\right)^{n+2} \geq \left(\frac{1}{27}\right)^{n-1}$$

A student produces the following argument:

$$\text{I:} \quad \left(\frac{1}{3}\right)^{n+2} \geq \left(\frac{1}{27}\right)^{n-1}$$

$$\text{II:} \quad \log_{\frac{1}{3}} \left(\left(\frac{1}{3}\right)^{n+2} \right) \geq \log_{\frac{1}{3}} \left(\left(\frac{1}{27}\right)^{n-1} \right)$$

$$\text{III:} \quad n+2 \geq (n-1) \log_{\frac{1}{3}} \left(\frac{1}{27} \right)$$

$$\text{IV:} \quad n+2 \geq 3(n-1)$$

$$\text{V:} \quad n+2 \geq 3n-3$$

$$\text{VI:} \quad n \leq 2$$

Which step (if any) in the argument is invalid?

- A There are no invalid steps; the argument is correct.
- B Only step **I** is invalid; the rest are correct.
- C Only step **II** is invalid; the rest are correct.
- D Only step **III** is invalid; the rest are correct.
- E Only step **IV** is invalid; the rest are correct.
- F Only step **V** is invalid; the rest are correct.
- G Only step **VI** is invalid; the rest are correct.

Trigonometry

9. The equation

$$2 \cos^2 x - k \sin x = 1$$

has n distinct solutions in the range $0 \leq x \leq 2\pi$

Consider the following statements:

I: $k = 0$ is **necessary** for $n = 2$

II: $k > 2$ **only if** $n = 0$

Which of the following statements is/are true?

A Neither of them

B I only

C II only

D I and II

Inequalities

10. A real number k has the property that

$$x^2 + kx + 1 \geq 0$$

for all real x satisfying $x^2 - 4x + 3 \leq 0$

Which of the following describes the set of all such values of k ?

A $k \geq -2$

B $k \geq -\frac{10}{3}$

C $-2 \leq k \leq 2$

D $-4 \leq k \leq 2$

E $-4 \leq k \leq 4$

F $-6 \leq k \leq -2$

G $k \geq 0$

11. Consider the curve defined by

$$(|x| - 4)^2 + (y - 1)^2 = 1$$

How many distinct straight lines passing through the origin are tangents to this curve?

A 0

B 1

C 2

D 3

E 4

F 5

12. Let a and b be positive real numbers, neither of which is equal to 1.

A student attempts to prove the following claim:

Claim: If $\log_a b = \log_b a$ then $a = b$

The student writes the following argument:

Line 1: Let $t = \log_a b$, so that $a^t = b$

Line 2: Since $b \neq 1$, we have $t \neq 0$

Line 3: Raising both sides of $a^t = b$ to the power $\frac{1}{t}$ gives $a = b^{1/t}$

Line 4: Hence $\log_b a = \frac{1}{t}$, so the given condition implies $t = \frac{1}{t}$, and therefore $t^2 = 1$

Line 5: Since $a^t = b$ and both a and b are positive, we must have $t > 0$

Line 6: Hence $t = 1$

Line 7: Therefore $b = a^1 = a$, which proves the claim.

Which line contains the first error?

A Line 1

B Line 2

C Line 3

D Line 4

E Line 5

F Line 6

G Line 7

H There is no error; the argument is correct.

Geometry

13. Let C be the circle $x^2 + y^2 = 9$

Let P be the point $(a, 0)$ where $a > 3$

The two tangents from P to C touch the circle at T_1 and T_2

What is the area of triangle PT_1T_2 in terms of a ?

A $3\sqrt{a^2 - 9}$

B $\frac{3(a^2 - 9)^{3/2}}{a^2}$

C $\frac{9}{a}\sqrt{a^2 - 9}$

D $\frac{a}{3}\sqrt{a^2 - 9}$

E $\frac{a^2 - 9}{3}$

F $\frac{a^2 - 9}{a}$

G $\frac{9}{\sqrt{a^2 - 9}}$

- 14.** Let $f(x) = |x - 1|$ and $g(x) = |x + 1|$

Define

$$h(x) = f(g(x))$$

How many integers x satisfy $h(x) < 1$?

A 0

B 1

C 2

D 3

E 4

F 5

Geometry

- 15.** A cuboid has edge lengths a , b and c . The diagonals of its three different rectangular faces have lengths 5 cm, $\sqrt{153}$ cm and $4\sqrt{10}$ cm.

What is the volume of the cuboid?

- A** 60 cm³
- B** 144 cm³
- C** 156 cm³
- D** 180 cm³
- E** 240 cm³
- F** 624 cm³

16. A student attempts to prove the following claim.

Claim: For every real number k with $k > \frac{2}{3}$ the equation $x^3 - 3k^2x + k = 0$ has three distinct real solutions.

The student produces the following argument:

- L1** Let $f(x) = x^3 - 3k^2x + k$, so $f'(x) = 3x^2 - 3k^2$, and $f'(x) = 0$ exactly when $x = -k$ or $x = k$, and these two values are different because $k \neq 0$
- L2** We have $f'(x) > 0$ for $x < -k$ and for $x > k$, and $f'(x) < 0$ for $-k < x < k$ so f has a local maximum at $x = -k$ and a local minimum at $x = k$
- L3** The local maximum value is $f(-k) = 2k^3 + k$, which is positive because $k > 0$
- L4** The local minimum value is $f(k) = k - 2k^3 < 0$
- L5** A cubic with positive x^3 coefficient whose local maximum value is positive and whose local minimum value is negative meets the x -axis at three different points, so the equation has three distinct real solutions

Which one of the following statements about this argument is correct?

- A** The first error is on line L1
- B** The first error is on line L2
- C** The first error is on line L3
- D** The first error is on line L4
- E** The first error is on line L5
- F** There is no error, and the claim is true.
- G** There is no error in any line, but the claim is false.

Polynomials

17. Consider the statement:

For all real numbers x , **if** $x^2 + px + 1 \leq 0$ **then** $x \leq 2$ (*)

What is the complete set of real values of p for which (*) is true?

A no real values of p

B all real values of p

C $p \geq 0$

D $p < 0$

E $p \geq -\frac{5}{2}$

F $p < -4$ **or** $p > 0$

18. A sequence x_n is defined by $x_1 = 1$ and

$$x_{n+1} = \frac{1}{3}x_n + 2$$

for $n \geq 1$.

Let S be the sum to infinity of the series

$$\sum_{n=1}^{\infty} (x_{n+1} - x_n)$$

What is the value of S ?

A $-\frac{3}{2}$

B -1

C 2

D $\frac{9}{2}$

E The series does not converge.

Number

- 19.** The highest common factor of two positive integers is 18, and their lowest common multiple is 3780.

What is the smallest possible value of the sum of the two integers?

- A** 504
- B** 522
- C** 540
- D** 558
- E** 666
- F** 738

Logic

20. Consider the statement about a real number k :

$$\text{For all real numbers } x \text{ there exists a real number } y \text{ such that} \quad (*)$$
$$(x - y)^2 + k(x + y) > 0$$

For which values of k is $(*)$ true?

- A** for no real values of k
- B** for all real values of k
- C** for all real k with $k > 0$
- D** for all real k with $k < 0$
- E** for all real k with $k \neq 0$
- F** for exactly one real value of k
- G** for all real k with $k \geq 0$
- H** for all real k with $k \leq 0$